EGI — Ethical Governance Intelligence

Hardware-Enforced AI Physical Safety IP Portfolio

To AI Safety Regulators and Industry Stakeholders,

EGI is a technical framework for an AI safety architecture that <u>physically enforces safety constraints after AI decision-making and before physical execution.</u>

Advances in AI robotics are making coexistence between humanity and AI a reality. In 2026, robot manufacturers across the globe have begun selling home-use AI robots, and AI robots are no longer the domain of science fiction.

Amid this transformation, concerns about AI are growing. Experts have noted that existing software-based safety measures are vulnerable to hacking, bugs, and jailbreaking.

While the EU, the State of California, China, and others are enacting AI safety legislation, **the EU AI Act will enter full force on 2 August 2026**. Yet the relevant industries have not yet established widely adopted solutions for preventing safety incidents involving AI robots. **A more robust AI safety technology framework is urgently required.**

EGI (Ethical Governance Intelligence) takes a hardware-based approach, building a physical safety technology system of **77 patents · 694 claims · 14 clusters (A–N)** anchored by a foundational Charter, and embodying an AI safety philosophy and conviction through **9 registered trademarks and 6 copyrights**.

The attached materials summarize the EGI Portfolio and are provided as reference. Out of the conviction that the coexistence of intelligent systems and humanity requires absolute safety measures that go beyond AI policy, EGI was developed; with that same conviction, **EGI Technical Compendium v6.0** is presented to the world.

May 2026

With deepest respect,

Jeong-Pyo Na, Developer, EGI, Republic of Korea

(Reference) EGI Portfolio — 77 Patents · 9 Trademarks · 6 Copyrights

EGI IP Portfolio (As of May 2026)

IP Type	Count	Status
Patent	**77**	Clusters A–N │ 694 Claims │ Filed at KIPO: 2026.1.30 – 2026.5.4.
Trademark	**9**	EGI · EGI-Standard · EGI-Gate · EGI-Safe · EGI-Certified · EGI-Net · EGI-Inside · HITA · Existence is Ethics
Copyright	**6**	EGI Charter C-2026-007803 │ Technical Compendium v1.0–v5.0 registered + v6.0 filing planned │ Automatic protection in 181 Berne Convention countries

EGI Patent Portfolio (77 Applications Filed, Clusters A–N)

Cluster	Count	Technology Domain	Filing Period	Key Regulatory Relevance
A	**10**	Invariant Condition-Based AI Governance Core (Intercept Module · Physical Isolation · Cryptographic Approval Enforcement. Axiomatic layer of the portfolio)	2026.01–02	EU AI Act Art. 9, 14, 15
B	**8**	Human Cognitive Safety and Interaction Control (Gaze · Spatial Pressure · Transparency · Non-Intervention Boundaries)	2026.02	EU AI Act Art. 50, Annex III; CCPA ADMT
C	**3**	AI Lifecycle Termination and Post-Control (Physical Memory Destruction Enforcement · Independent Circuit Self-Neutralization)	2026.02	EU AI Act Art. 17; GDPR Art. 17; CCPA Disposal Provisions
D	**9**	Quantum-Resistant Invariant Condition Security (Pre-emptive defense for the post-quantum era)	2026.02–03	NIST PQC; EU NIS2; US NDAA
E	**8**	Power Path and Execution Route Physical Cutoff (Zero-Current Transition Lock · Irreversible Wire Break · Active Energy Erasure. Lowest physical layer)	2026.03	ISO 13849-1 · IEC 60204-1 · IEC 61508 · EU Machinery Directive
F	**4**	Humanoid Robot Physical Safety Control (Biosignal Torque Attenuation · Joint Brake Engagement · Collision Energy Limiting · Counter-Torque)	2026.03	ISO 13482 · ISO/TS 15066 · EU AI Act Annex III
G	**3**	State Transition and Recovery Control (Vulnerable Entity Protection Locomotion Transition · Spatial Condition Auto-Restoration · Heartbeat Fail-Safe)	2026.04	ISO 13849 PL e · IEC 61508 SIL 3
H	**3**	Multi-Sensor Consistency · Quorum · Irreversible Recording (WORM Raw Value Storage · 3-Axis Consistency Enforcement)	2026.04	EU AI Act Art. 12, 15; ISO 26262 ASIL D
I	**3**	Network and Command Boundary Physical Verification (Cloud-Edge Verification · Network Disconnection Fail-Safe)	2026.04	EU NIS2 · CRA · UN R155/R156
J	**6**	Swarm Robot Safety Control (Invariant Condition Enforcement · Defector Physical Isolation · Swarm Emergent Risk Blocking. First-of-its-kind worldwide)	2026.04	EU AI Act Annex III · NIST AI RMF
K	**5**	Vulnerable Entity Protection — 5-Axis Orthogonal (Linguistic Context · Spatial OR/AND · Kinematics · Dynamics) Single and Dual Blocking Structures	2026.04–05	ISO 13482 · EU AI Act Annex III · UN CRPD · UN CRC

Cluster	Count	Technology Domain	Filing Period	Key Regulatory Relevance
L	**5**	Learner Protection — Imitation Behavior · Group Dynamics (3-Tier Asymmetry) · Supervisor Approval · Arts/PE Postures · Hazardous Teaching Aids	2026.04–05	EU AI Act Annex III (Education) · UN CRC · UNESCO AI Ethics
M	**5**	Medical AI Robot Safety Control — Anatomical Protection Zones · Physical Overload · Vital Signal Maintenance · Sole Surgeon Approval (4-Dimensional Safety Range)	2026.04–05	EU MDR 2017/745 · EU AI Act Annex III · FDA 510(k)/PMA · IEC 60601-1
N	**5**	Care Robot Safety Control — Cumulative Interaction · Dual Intent Consistency · Harm Threshold Notification · Autonomy Domain Non-Intervention	2026.04–05	WHO Healthy Ageing Strategy · EU AI Act Annex III · UN Principles for Older Persons · ISO 13482
Total: 77 Patents ∣ 694 Claims ∣ 14 Clusters ∣ KIPO Filing: 2026.1.30 – 2026.5.4 ∣ PCT International Filing Planned (within priority period)				